

SEO Agency Platform: JavaScript-Only Features, Use Cases, and 10-Day Timeline

1. Project Scope

This document defines a 10-day company-level foundation plan for an SEO agency platform built completely in JavaScript.

The product must use JavaScript across the full stack:

- Frontend: React.js.
- Backend/API: Node.js with Express.js or NestJS.
- Database access: Prisma or Sequelize from Node.js.
- Jobs/workers: Node.js worker process with BullMQ or a similar JS queue.
- PDF generation: Node.js with Playwright/Puppeteer.
- Crawling/audits: Node.js crawler services.
- AI/integrations: Node.js adapter services.

This is not a throwaway demo plan. The 10-day target is a company-level MVP foundation that can grow into production. The first release must prove the full operating loop:

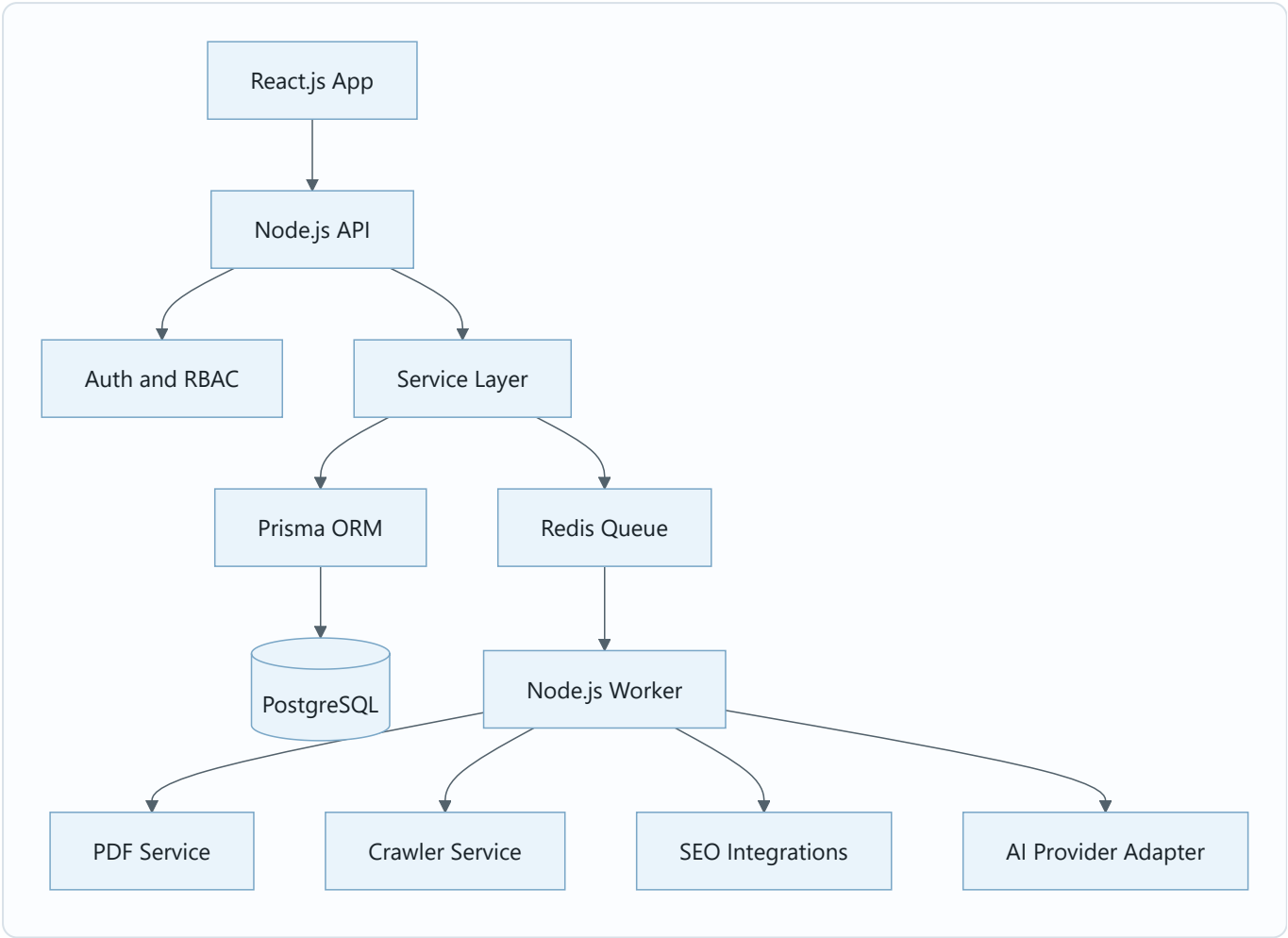
Client -> Website -> SEO Data -> Insight -> Task -> Report -> Client Portal

2. Company-Level Architecture

2.1 Stack

Layer	Technology
Frontend	React.js, Vite, React Router, React Query
Styling	Tailwind CSS or structured CSS modules
Charts	Recharts
Backend	Node.js API using Express.js or NestJS
Database	PostgreSQL
ORM	Prisma recommended
Auth	JWT access token plus secure refresh token
Jobs	Node.js worker plus BullMQ/Redis
PDF	Playwright or Puppeteer from Node.js
Crawling	Node.js crawler service
AI	Node.js LLM adapter layer
Integrations	Node.js provider adapters for GSC, GA4, PageSpeed, Semrush, Ahrefs later
Testing	Vitest/Jest, React Testing Library, Supertest, Playwright
Deployment	Node.js hosting, PostgreSQL, Redis, object storage

2.2 Architecture Diagram



3. Core Features

3.1 User, Role, and Access Management

- Agency workspace.
- User login and signup.
- JWT authentication.
- Refresh-token handling.
- Role-based access control.
- Roles: Admin, Manager, Employee, Client.
- Client assignment rules.
- Protected React routes.
- Backend route guards in Node.js.
- Tenant isolation by agency and client.
- Client-safe response serializers.

3.2 Client Management

- Add, edit, view, and delete clients.
- Store client company profile.

- Store client contacts.
- Add internal notes.
- Track client status: active, paused, churned.
- Store retainer data for internal users only.
- Assign managers and employees to clients.
- View client activity history.
- Client health score foundation.

3.3 Website Management

- Add, edit, view, and delete websites.
- Link websites to clients.
- Store domain, status, SEO score, CMS, target locations, business category, preferred locale, and search engine target.
- Add competitors for each website.
- Website workspace with tabs: Overview, Keywords, Audits, Competitors, Backlinks, Content, Tasks, Reports.

3.4 SEO Dashboard

- SEO score overview.
- Organic traffic trend.
- Clicks, impressions, CTR, and average position.
- Keyword movement summary.
- Top ranking keywords.
- Ranking distribution.
- Audit issue summary.
- Backlink summary.
- Task summary.
- Mock/live data indicator.
- Client-safe dashboard version.

3.5 Keyword Research and Keyword Management

- Discover keywords by seed term, URL, competitor domain, service category, and location.
- Generate keyword ideas grouped by questions, comparisons, prepositions, related terms, and long-tail opportunities.
- Store search volume, difficulty, CPC, intent, target page, device, location, language, and SERP feature presence.
- Add and edit tracked keywords.
- Group keywords by campaign, topic cluster, funnel stage, location, or client priority.
- Track ranking history over time.
- Show improved, declined, unchanged, top 3, top 10, and top 100 keywords.
- Identify keyword gaps against competitors.
- Identify keyword cannibalization where multiple client pages compete for the same query.
- Calculate keyword opportunity score using volume, difficulty, ranking distance, business value, and current page strength.
- Create content briefs or SEO tasks directly from keyword opportunities.
- CSV import/export.

3.6 Competitor Intelligence

- Add competitors per website.
- Compare client keywords against competitors.

- Identify missing keywords.
- Identify shared keywords.
- Show competitor top pages.
- Show competitor visibility score.
- Track competitor content velocity.
- Track competitor backlink sources.
- Identify competitor pages that should be outranked or used as content references.
- Basic share-of-voice calculation.

3.7 Backlink Research and Link Building

- Store backlink inventory for each website.
- Track referring domains, source URLs, target URLs, anchor text, dofollow/nofollow status, and authority metrics.
- Identify new and lost backlinks.
- Identify broken backlinks pointing to missing client pages.
- Identify toxic or suspicious backlinks for review.
- Compare backlinks against competitors.
- Run backlink gap/link intersect analysis to find domains linking to competitors but not the client.
- Create link-building prospects from competitor backlink sources.
- Track outreach status without turning the product into a full CRM.
- Convert backlink opportunities or broken backlink issues into tasks.
- Include backlink wins/losses and link-building progress in reports.

3.8 Technical SEO Audit

- Trigger website audit from React UI.
- Queue audit job in Node.js worker.
- Store audit runs.
- Store crawled URL data.
- Generate issue list.
- Detect basic technical SEO issues:
 - Broken links.
 - Missing title.
 - Duplicate title.
 - Missing meta description.
 - Duplicate meta description.
 - Missing H1.
 - Multiple H1 tags.
 - Image alt text issues.
 - Noindex pages.
 - Redirect chains.
 - Canonical issues.
 - Sitemap gaps.
 - Robots.txt blocks.
 - Slow pages.
- Assign severity: critical, high, medium, low.
- Mark issues as resolved.
- Convert audit issue into task.

3.9 Task and Workflow Management

- Create, edit, view, and delete tasks.
- Assign tasks to users.
- Link tasks to clients, websites, keywords, audit issues, or reports.
- Set category, priority, status, and deadline.
- Add comments.
- Add attachment metadata.
- View task history.
- Employee My Tasks dashboard.
- Manager workload view.
- Activity logs.

3.10 Reporting and Client Portal

- Generate monthly SEO reports.
- Report sections:
 - Executive summary.
 - Traffic performance.
 - Keyword wins and losses.
 - Technical issues fixed.
 - Open technical issues.
 - Completed tasks.
 - Content work.
 - Backlink summary.
 - Next-month plan.
- PDF export using Node.js Playwright/Puppeteer.
- Report approval flow.
- Client portal report view.
- Client dashboard with restricted fields.

3.11 AI Assistant

- Ranking drop explanation.
- Keyword suggestions.
- Content brief generation.
- Performance summary.
- Audit issue summary.
- Issue-to-task draft creation.
- Store AI request history.
- Apply rate limiting on AI endpoints.
- Keep AI output internal until approved.

3.12 Alerts and Notifications

- Task assigned notification.
- Deadline approaching notification.
- Ranking drop alert.
- Traffic drop alert.

- Critical audit issue alert.
- Report ready alert.
- Integration sync failure alert.

3.13 Integrations Foundation

- Mock integration provider for development.
- JavaScript adapter interface for real integrations.
- Future integration targets:
 - Google Search Console.
 - Google Analytics 4.
 - PageSpeed Insights.
 - Google Business Profile.
 - Semrush.
 - Ahrefs.

3.14 AI Platform Research and Optimization

This platform should not only use AI to write text. It should use AI platforms as an SEO research and optimization layer.

The AI workflow works like this:

1. Collect structured SEO data from the platform: keywords, rankings, pages, audits, competitors, backlinks, traffic, tasks, and reports.
2. Normalize that data in the Node.js service layer so the AI receives clean context instead of raw database dumps.
3. Send scoped prompts to the AI provider through a JavaScript adapter.
4. Generate research outputs: keyword opportunities, topic clusters, content gaps, backlink prospects, competitor summaries, and technical issue explanations.
5. Generate optimization outputs: title/meta suggestions, content brief recommendations, internal link suggestions, schema recommendations, page refresh guidance, and report summaries.
6. Store every AI request and result with user, client, input context, output, provider, and timestamp.
7. Keep AI output internal by default. A manager must approve anything before it appears in a client report or portal.

AI platform features:

- Keyword research expansion from seed keywords, competitor domains, and page URLs.
- Content gap analysis using competitor pages and missing keyword clusters.
- Backlink prospect prioritization based on competitor links and topical relevance.
- Technical audit explanation in plain language.
- On-page optimization recommendations.
- Content brief generation.
- Existing content refresh recommendations.
- Internal linking recommendations.
- Report executive summaries.
- AI search/AEO readiness checks for future platforms like AI Overviews, ChatGPT, Gemini, and Perplexity-style answer engines.

4. Primary Actors

Actor	Description
Admin	Agency owner or administrator with full access.
Manager	SEO manager responsible for assigned clients and reports.
Employee	SEO specialist responsible for assigned execution tasks.
Client	External client with restricted portal access.
Node.js Worker	Background process for crawls, reports, alerts, syncs, and AI jobs.
Integration Provider	External data source such as GSC, GA4, Semrush, Ahrefs, or PageSpeed.
AI Provider	LLM provider used for summaries, briefs, and recommendations.

5. Use Cases

5.1 Admin Use Cases

- Sign up and create agency workspace.
- Invite users.
- Assign roles.
- Add clients.
- Add client contacts.
- Add websites.
- Assign team members to clients.
- View all agency KPIs.
- Manage integrations.
- Review all reports.
- Configure white-label branding.

5.2 Manager Use Cases

- View assigned clients.
- View website SEO dashboard.
- Add or update keywords.
- Review ranking movement.
- Run website audit.
- Convert audit issue into task.
- Assign tasks to employees.
- Review team workload.
- Generate monthly report.
- Approve report for client.
- Use AI assistant for strategy support.

5.3 Employee Use Cases

- View assigned tasks.
- Open task detail.

- Review client and website context.
- Add task comments.
- Upload task attachment metadata.
- Update task status.
- Use AI assistant to create briefs or summaries.
- Mark task complete.

5.4 Client Use Cases

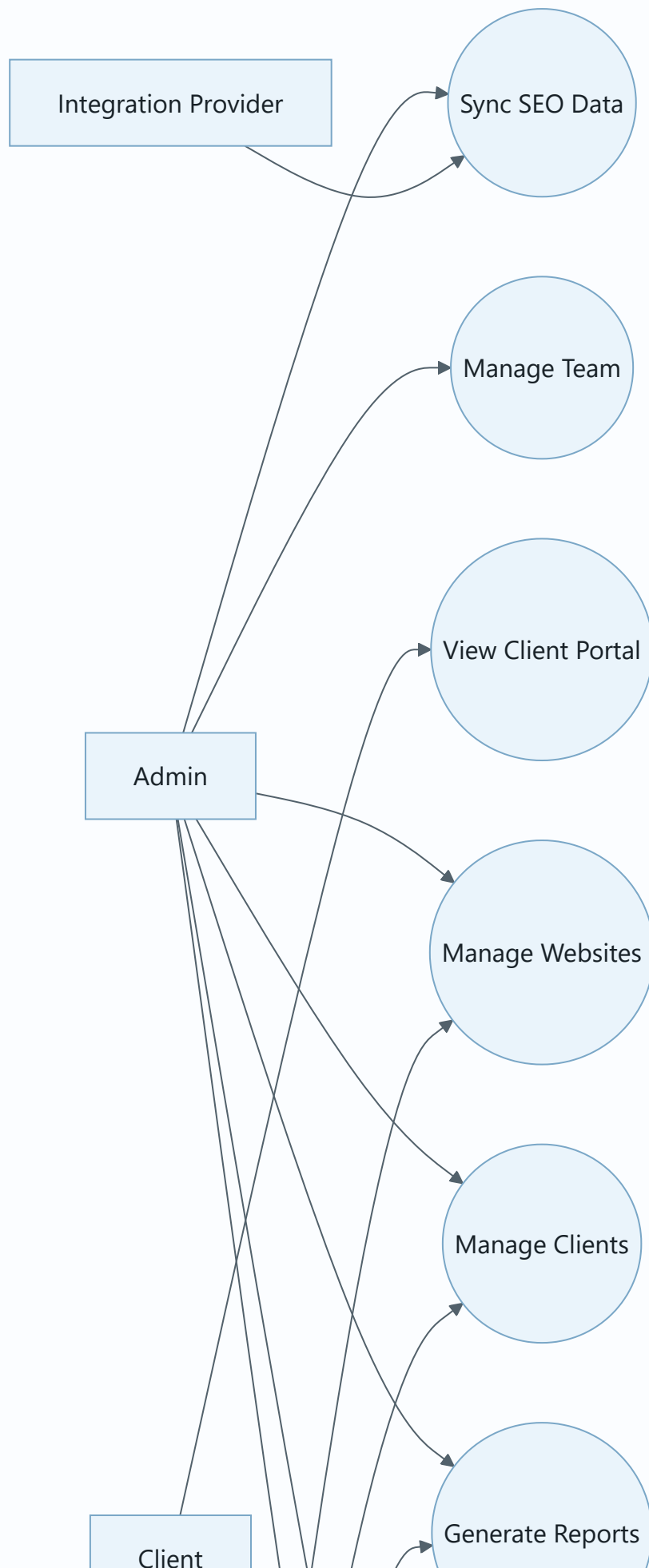
- Log in to client portal.
- View client-safe SEO dashboard.
- View completed work.
- View current priorities.
- Download approved reports.
- View recommendations.

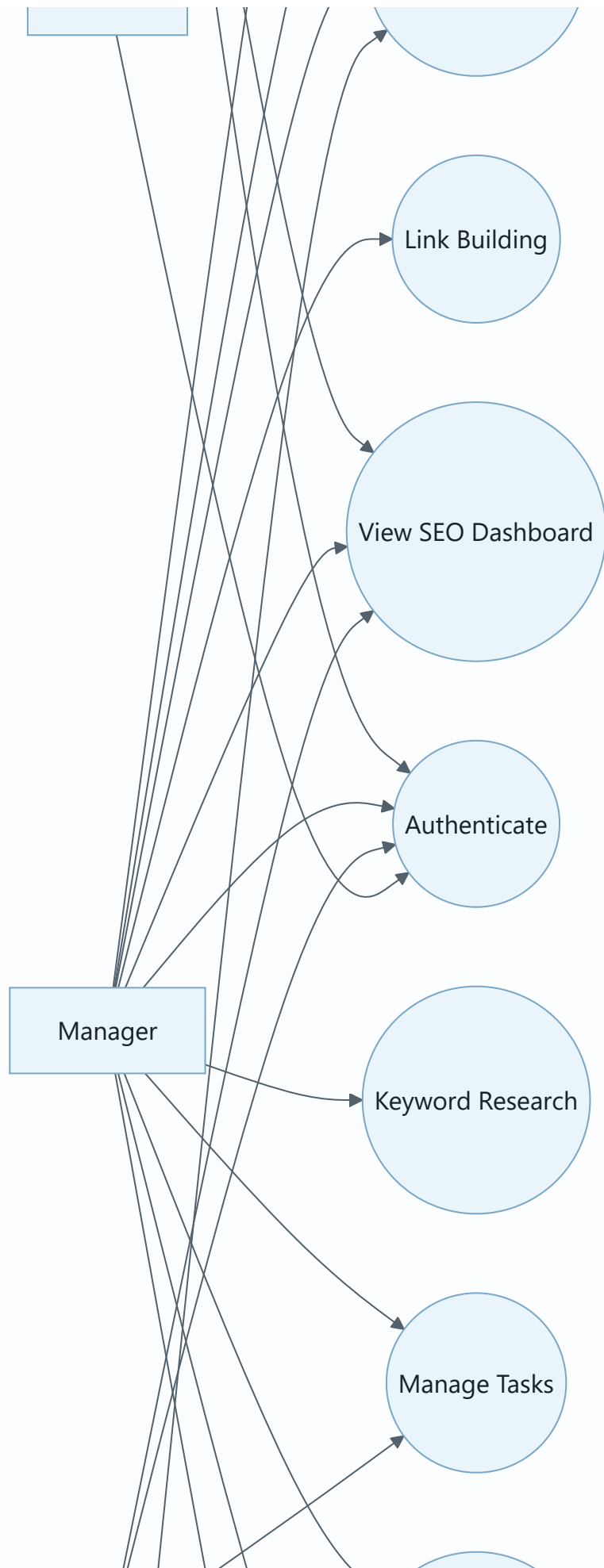
5.5 System Use Cases

- Sync SEO data from mock/live providers.
- Generate keyword ranking history.
- Run technical audit in Node.js worker.
- Generate alerts.
- Generate PDF report from Node.js.
- Store activity logs.
- Enforce tenant isolation in backend route guards and service layer.

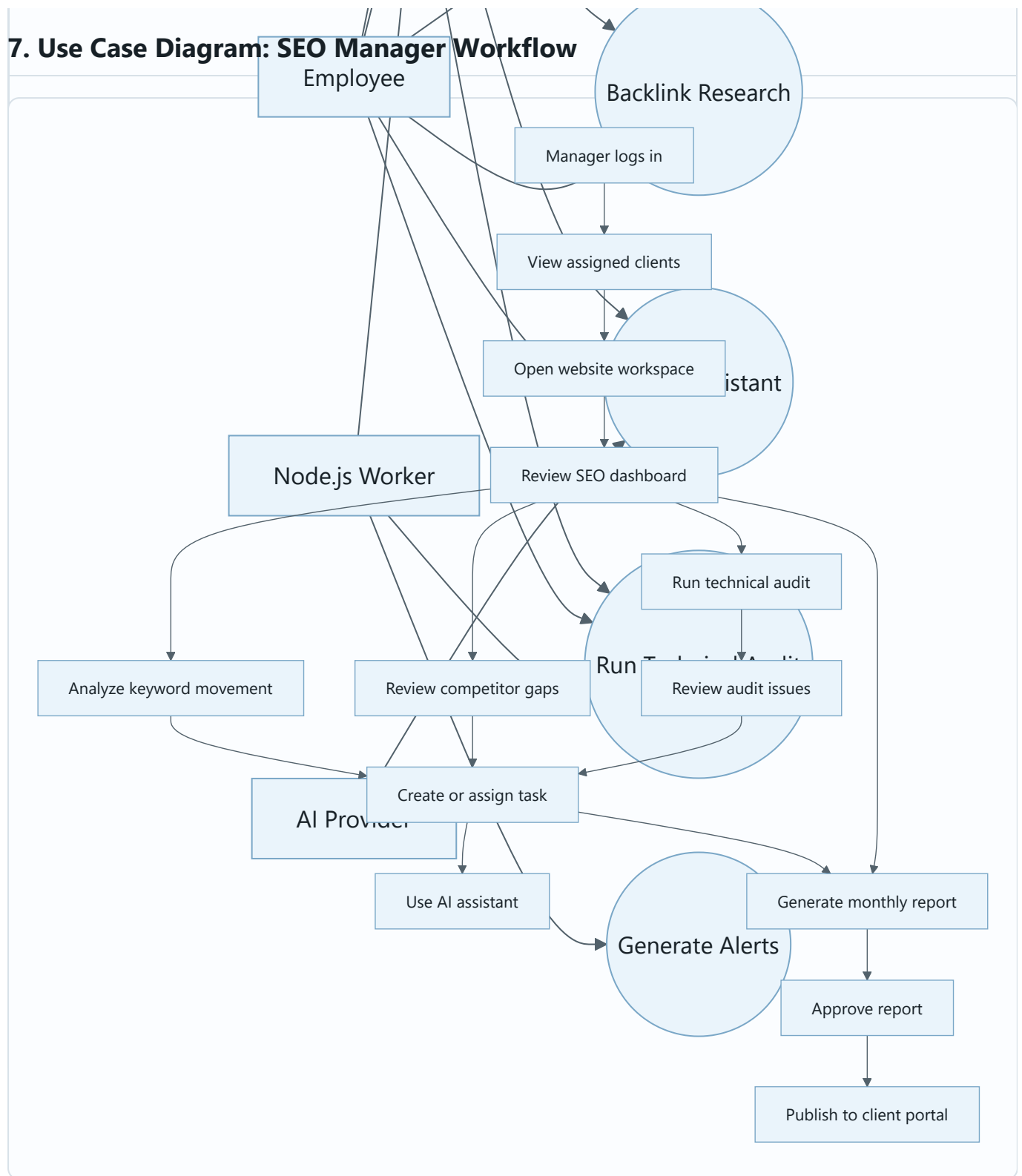
6. Use Case Diagram: Overall System



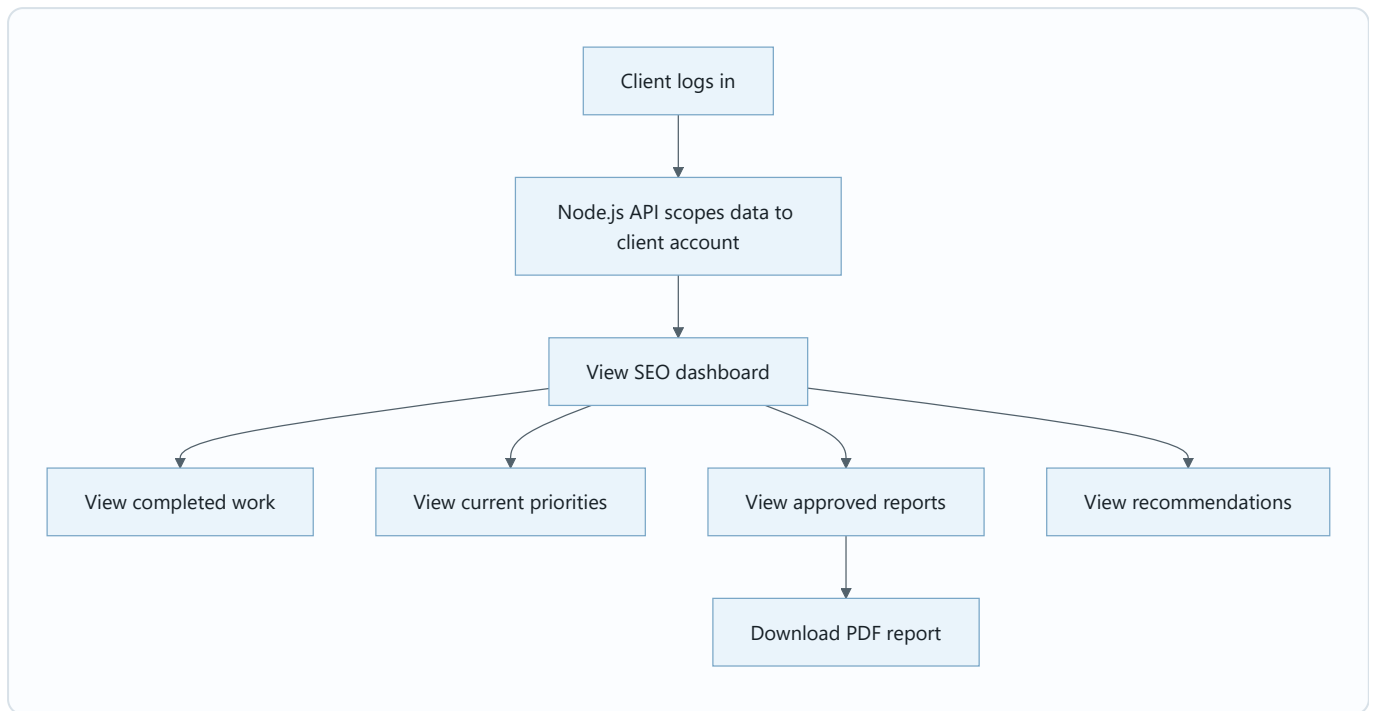




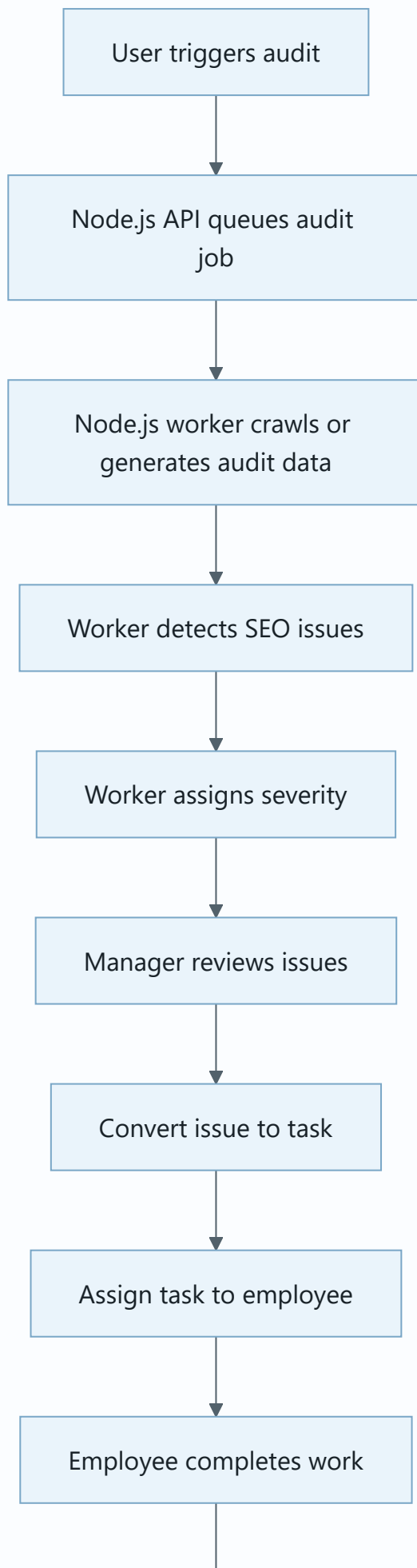
7. Use Case Diagram: SEO Manager Workflow



8. Use Case Diagram: Client Portal



9. Use Case Diagram: Audit-to-Task Flow

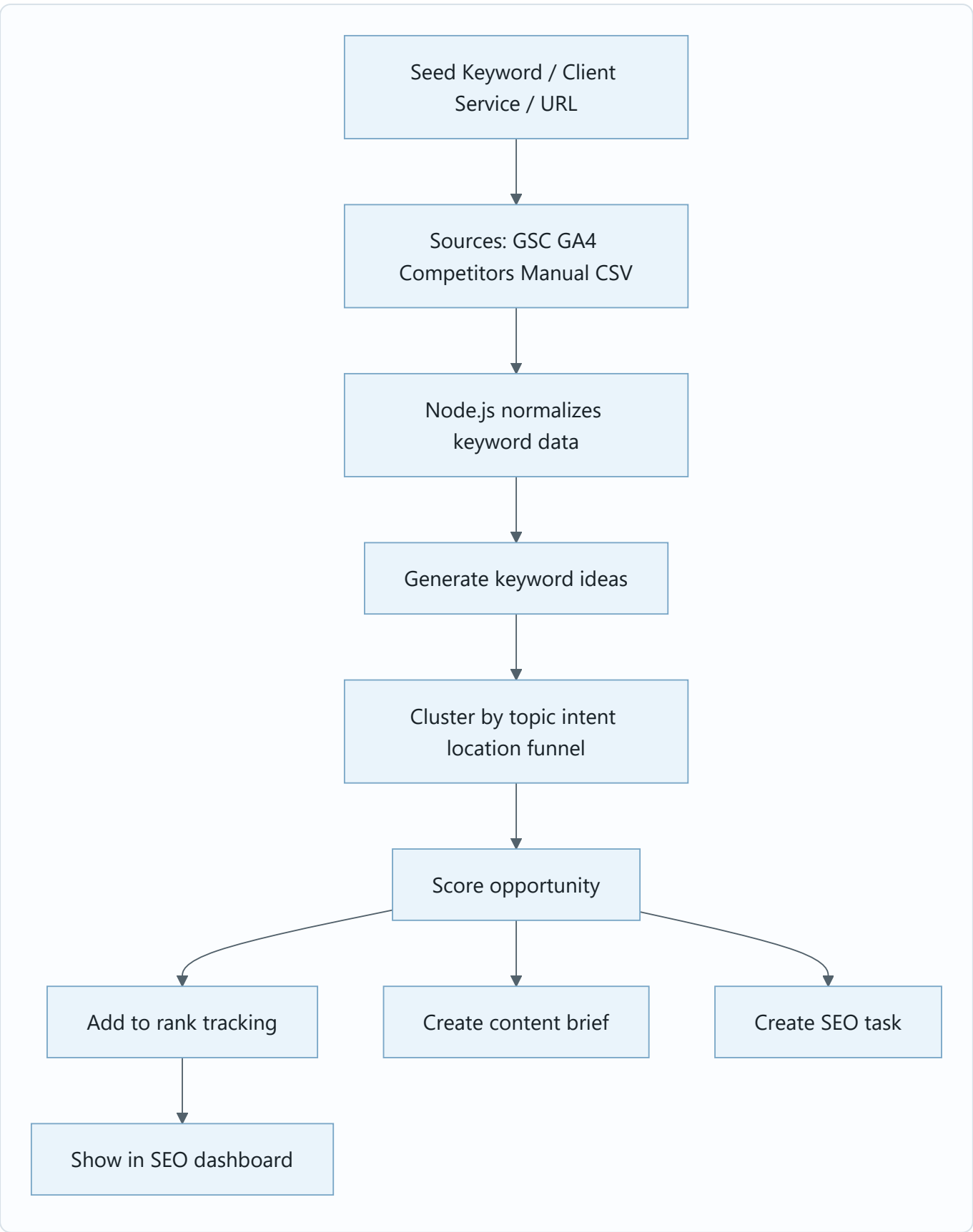


Issue marked resolved

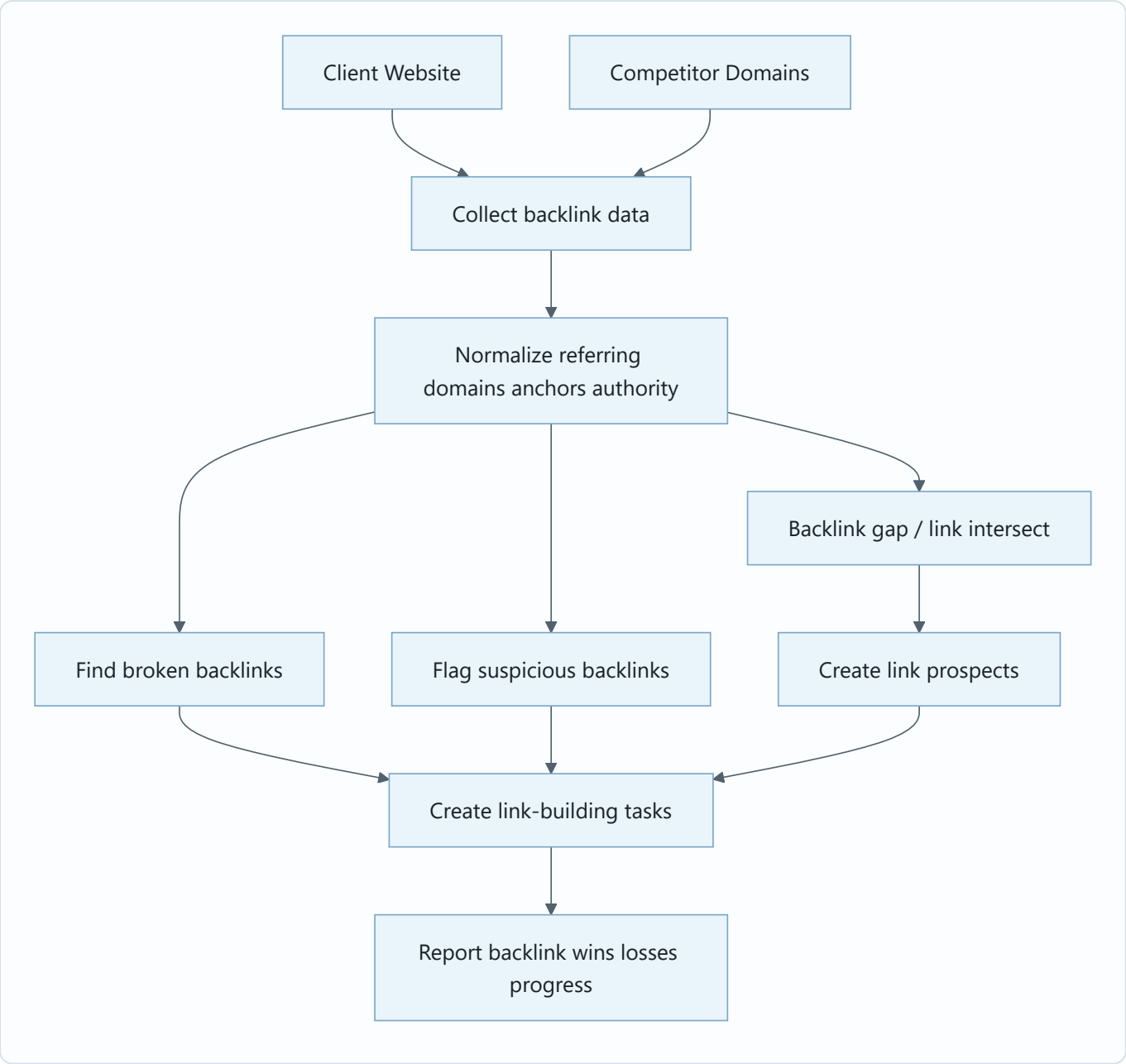


Completed work appears in
report

10. Use Case Diagram: Keyword Research

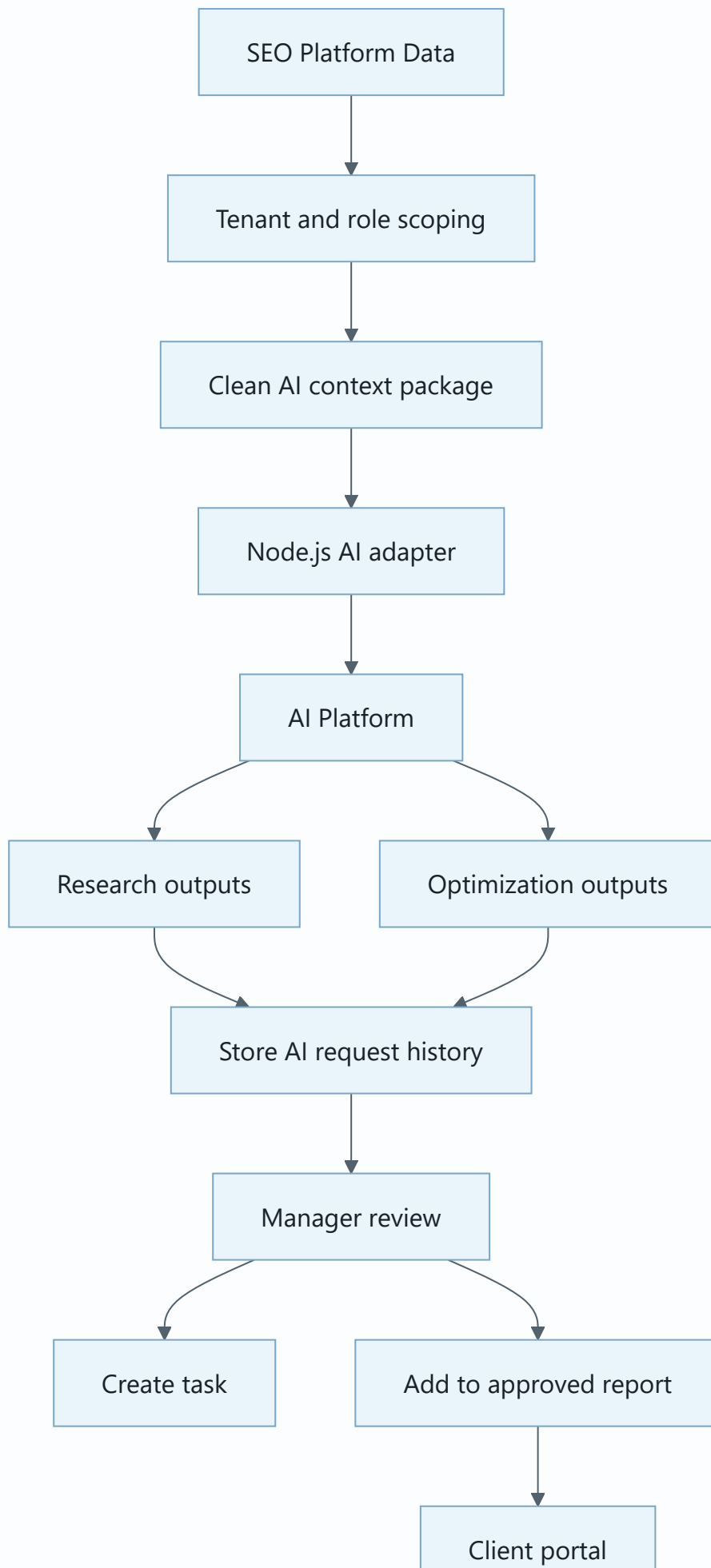


11. Use Case Diagram: Backlink Research and Link Building

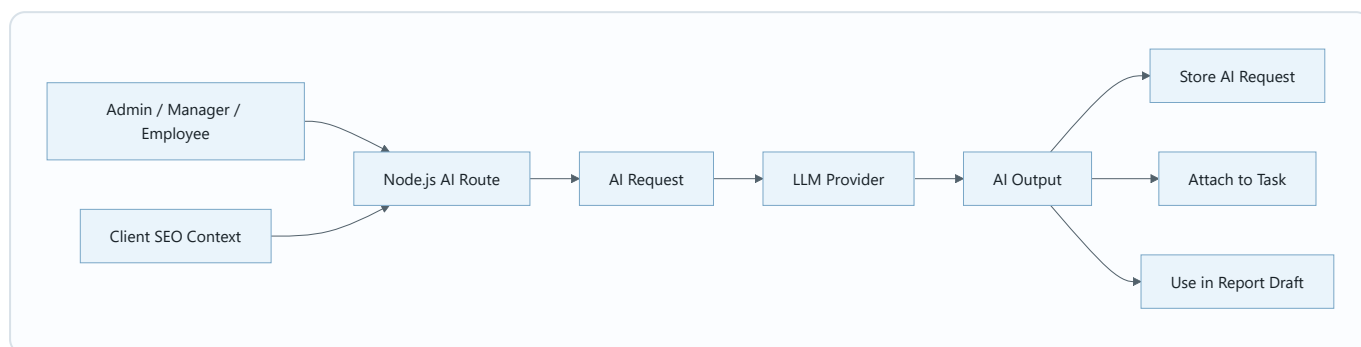


12. Use Case Diagram: AI Research and Optimization Flow





13. Use Case Diagram: AI Assistant



14. 10-Day Company-Level JavaScript Timeline

Day 1: JavaScript Project Foundation

Focus: Establish the full-stack JavaScript architecture.

- Set up React.js app with Vite.
- Set up Node.js backend API project.
- Set up shared JavaScript config and environment handling.
- Set up PostgreSQL connection through Prisma.
- Define Prisma schema for agencies, users, roles, clients, websites, competitors, keywords, audits, tasks, reports, alerts, integrations, and AI requests.
- Define RBAC permissions in JavaScript.
- Define tenant scoping helpers.
- Define client-safe serializers.
- Add lint/build/test scripts.

Output: Full-stack JavaScript foundation ready for feature implementation.

Day 2: Auth, Roles, and Tenant Security

Focus: Build secure access control first.

- Implement signup and login in Node.js.
- Hash passwords with a Node.js library.
- Issue JWT access tokens.
- Add refresh-token flow.
- Add Admin, Manager, Employee, and Client roles.
- Add backend route guards.
- Add protected React routes.
- Add tenant-isolation tests.

Output: Users can authenticate and access only allowed data.

Day 3: Client and Team Management

Focus: Build agency operating foundation.

- Build client CRUD API.
- Build client list/detail UI.
- Build contacts and internal notes.
- Add team user management.
- Assign managers and employees to clients.
- Add activity log entries.
- Hide retainer/internal fields from client role.

Output: Agency users can manage clients and team assignments safely.

Day 4: Website and Competitor Management

Focus: Make websites the center of SEO work.

- Build website CRUD API and UI.
- Store CMS, business category, locations, locale, and search engines.
- Build website workspace tabs.
- Add competitor CRUD API and UI.
- Add backlink research data model.
- Add competitor backlink source placeholders.
- Add seed/demo data through JS seed script.

Output: Each client can have websites, competitors, and project-level SEO settings.

Day 5: Keyword Tracking and Ranking History

Focus: Build keyword intelligence foundation.

- Build keyword CRUD API and UI.
- Build keyword groups.
- Add ranking history model.
- Add mock ranking generator in Node.js.
- Add keyword table filters.
- Add ranking trend charts.
- Add keyword research workflow: seed keywords, competitor keywords, questions, related terms, and long-tail ideas.
- Add keyword opportunity scoring.
- Add import/export foundation.

Output: Users can track keywords and ranking movement.

Day 6: SEO Dashboard and Share of Voice

Focus: Build the main performance dashboard.

- Build SEO summary API.
- Add traffic trend API.
- Add KPI cards in React.
- Add ranking distribution chart.
- Add keyword movement chart.
- Add top pages section.
- Add competitor visibility/share-of-voice calculation.

- Add mock/live labels.
- Add client-safe dashboard response.

Output: Internal users and clients can view scoped SEO performance.

Day 7: Technical Audit Engine V1

Focus: Build Node.js crawl/audit foundation.

- Add audit run model.
- Add crawled URL model.
- Add technical issue model.
- Create Node.js audit worker.
- Generate rule-based technical issues.
- Add issue severity scoring.
- Build audit UI.
- Add mark-resolved action.
- Add issue-to-task conversion.

Output: Users can run audits, inspect issues, and convert issues into work.

Day 8: Task Workflow and Alerts

Focus: Connect SEO insights to execution.

- Build task CRUD API and UI.
- Add comments and attachment metadata.
- Add assignment flow.
- Add status transitions.
- Add My Tasks view.
- Add manager workload view.
- Add notifications and alerts.
- Add activity logging.

Output: SEO findings can become assigned and trackable work.

Day 9: Reports and Client Portal

Focus: Package work into client-facing outcomes.

- Build report model and API.
- Aggregate traffic, keyword, audit, task, and recommendation data.
- Build report preview UI.
- Generate PDF with Node.js Playwright/Puppeteer.
- Add approval workflow.
- Build client portal dashboard.
- Build approved reports page.

Output: Clients can view approved progress and download reports.

Day 10: AI, QA, Hardening, and Deployment Prep

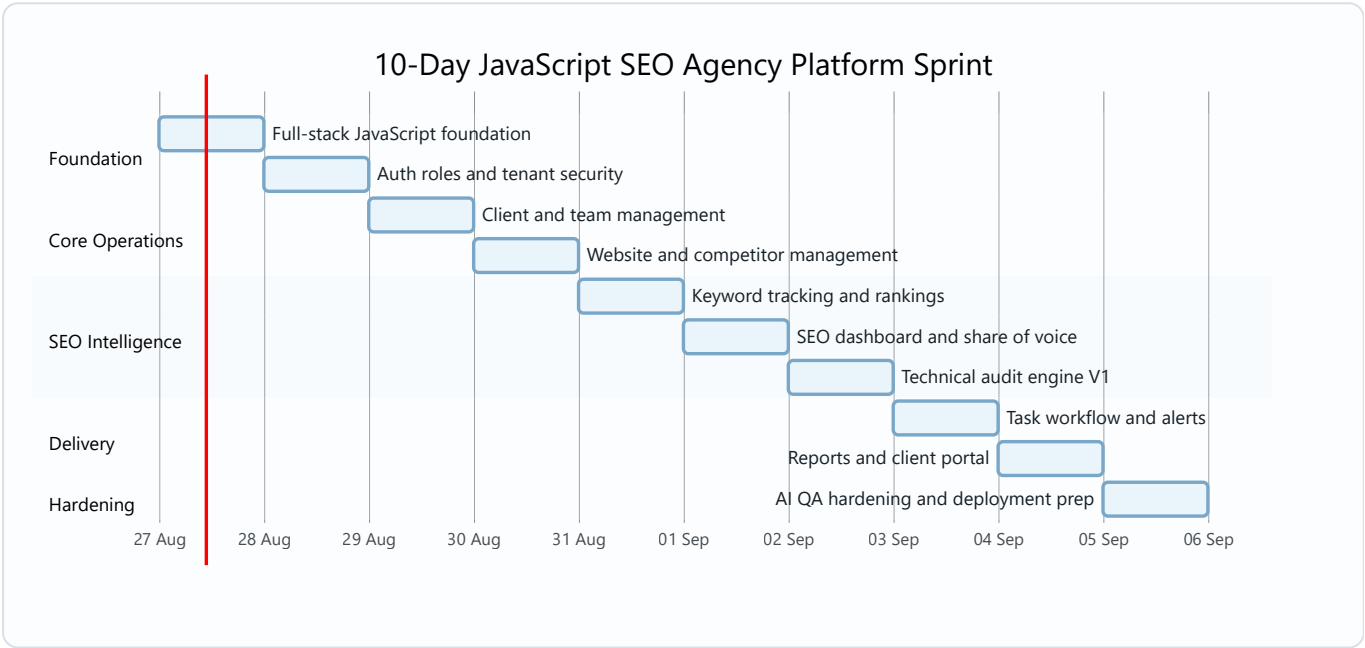
Focus: Make the MVP foundation credible for company use.

- Add Node.js AI provider abstraction.

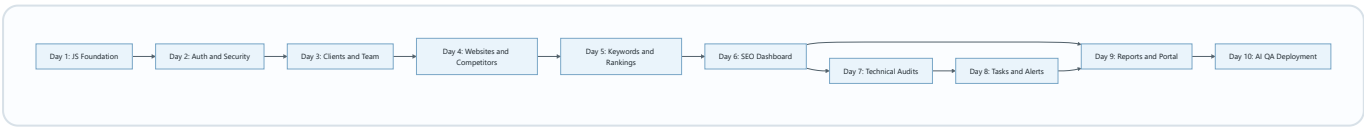
- Add prompts for keyword research expansion, backlink prospect research, ranking explanations, content briefs, audit summaries, and report summaries.
- Store AI request history.
- Add rate limiting.
- Run frontend tests.
- Run backend API tests.
- Run tenant-isolation tests.
- Run lint/build checks.
- Prepare deployment checklist.
- Fix critical security and UX issues.

Output: Company-level JavaScript MVP foundation ready for staging.

15. Timeline Diagram



16. Dependency Timeline



17. Expected End-of-Sprint Flow

At the end of 10 days, the product should support this company-level MVP flow:

1. Admin creates agency workspace.
2. Admin invites team members.
3. Admin or Manager creates a client.
4. Manager adds websites and competitors.
5. Manager tracks keywords and ranking history.
6. Manager reviews SEO dashboard.
7. Manager runs keyword research and adds priority keyword opportunities.
8. Manager reviews backlink gaps and creates link-building prospects.

9. Manager runs a technical audit.
10. Manager converts an issue or backlink opportunity into a task.
11. Employee completes the assigned task.
12. AI assistant generates an internal optimization summary.
13. Manager reviews and approves the AI-supported recommendation.
14. Manager generates and approves a report.
15. Client logs in and views the approved report and client-safe dashboard.

18. Out of Scope for the First 10 Days

- Billing/subscription system.
- Full CRM/sales pipeline.
- Enterprise SSO.
- Custom white-label domains.
- Production-scale backlink index.
- Production-scale crawler cluster.
- Advanced AI visibility tracking.
- Fully automated AI agents.

These are not rejected features. They are later-phase work after the JavaScript MVP foundation is stable.

19. Key Constraint

Because this project must be fully JavaScript, no Python, FastAPI, Django, Flask, or Python-based PDF tooling should be used.

All implementation should use JavaScript or Node.js equivalents.